[fn10] The set of events at a space-like distance from a reference event.

[fn11] The astute reader will object that the *halfway* moment of my quarter of an hour can be considered simultaneous to your reply. The reader who has studied physics will recognize that this is ‘Einstein’s convention’ for defining simultaneity. This definition of simultaneity depends on how I move, and consequently does not define simultaneity between two events but only a simultaneity *relative* to the state of movement of particular bodies. In figure 3.3 a dot is halfway between *a* and *b*, the points at which I exit from the past of the observer and enter his future. The other dot is halfway between *e* and *d*, the points at which I exit from the past of the observer and enter into his future if I move along a different trajectory. Both dots are simultaneous as regards the reader, according to this definition of ‘simultaneity’, but they occur in successive times. The two dots are each simultaneous to the reader, but relative to two different motions of mine. Hence the term ‘relativity’.

[fn12] Airplane and ball follow a geodesic in a curved space. In the case of the ball, the geometry is approximately given by the metric $ds^2 = (1 - 2\Phi(x))\, dt^2 - dx^2$, where $\Phi(x)$ is the Newtonian potential. The effect of the gravitational field is reduced to the dilation of time with altitude. (The reader familiar with the theory will notice the curious sign inversion: the physical trajectory maximizes proper time.)

[fn13] Observations of the binary system PSR B193+16 show that the two stars which revolve around one another radiate gravitational waves. These observations brought a Nobel Prize for Russell Hulse and Joseph Taylor in 1993.

[fn14] Plutarch, *Adversus colotem*, 4, 1108. The word φύσιν means ‘nature’, and includes the sense ‘the nature of something’.

[fn15] This term is called ‘cosmological’ because its effects occur only at an extremely large, or ‘cosmological’ distances. The constant Λ is called the ‘cosmological constant’, and its value was measured at the end of the 1990s, bringing a Nobel Prize in 2011 for the astronomers Saul Perlmutter, Brian P. Schmidt and Adam G. Riess.

[fn16] Göttingen, where Hilbert worked, was at this time the seat of the most important school of geometry.

[fn17] A sphere is the set of points in R^3 determined by the equation $x^2 + y^2 + z^2 = 1$. The 3-sphere is the set of points in R^4 determined by the equation $x^2 + y^2 + z^2 + u^2 = 1$.

[fn18] It has been objected that Dante speaks of ‘circles’ and not of ‘spheres’. But the objection is invalid. Brunetto Latini writes of ‘a circle, like the shell of an egg’. The word ‘circle’, for Dante, as for his teacher and mentor, designates everything which is circular, including spheres.